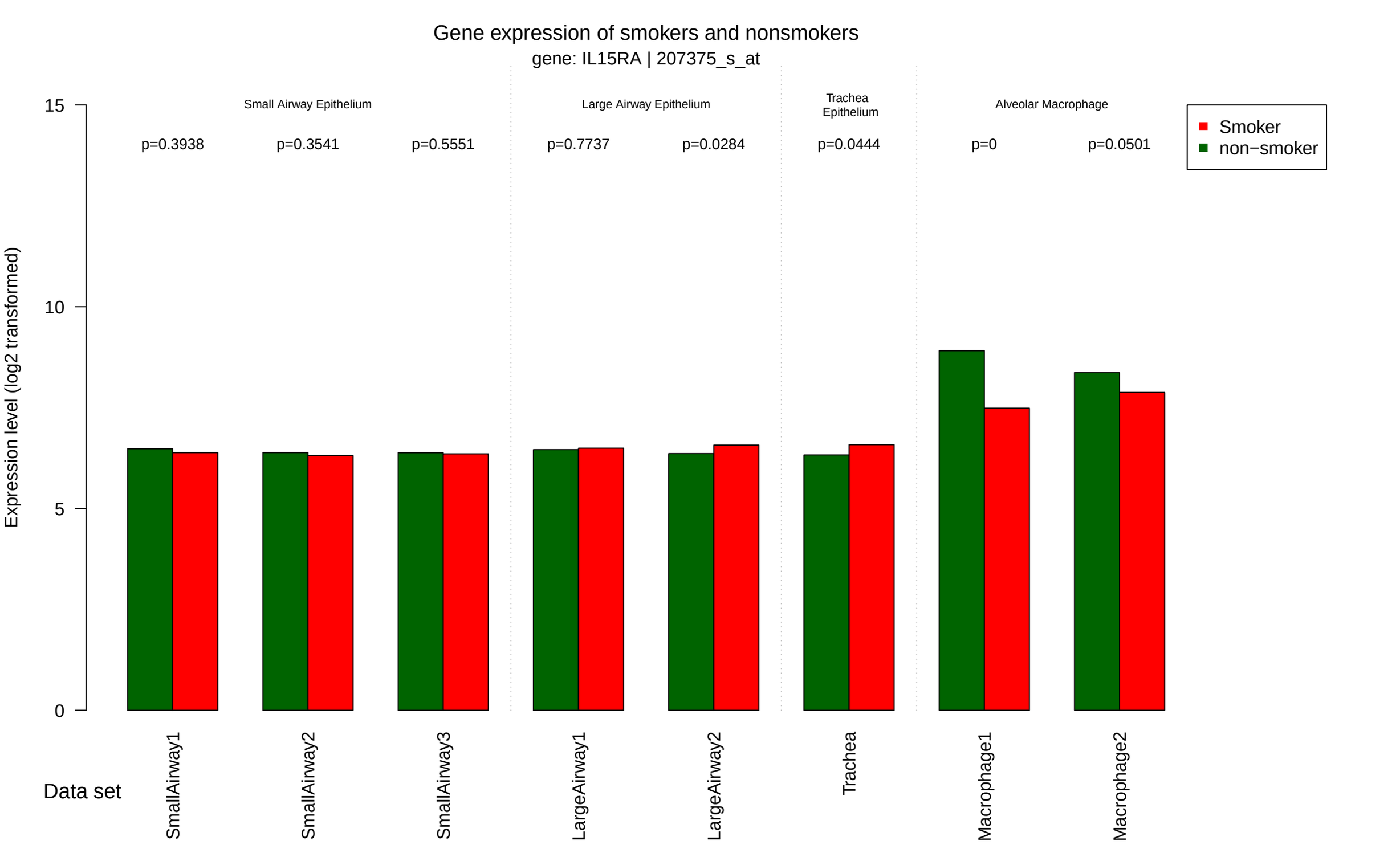
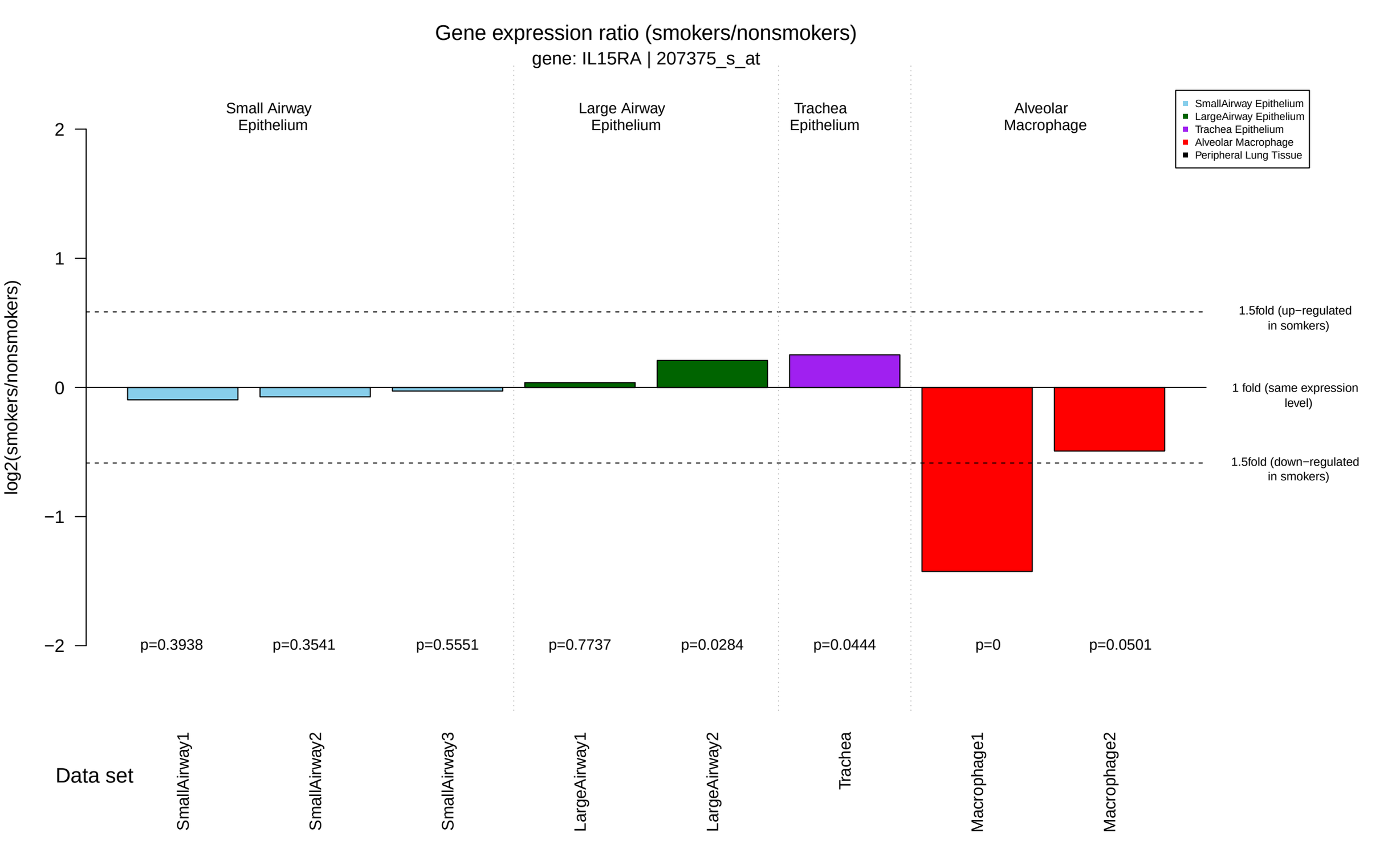
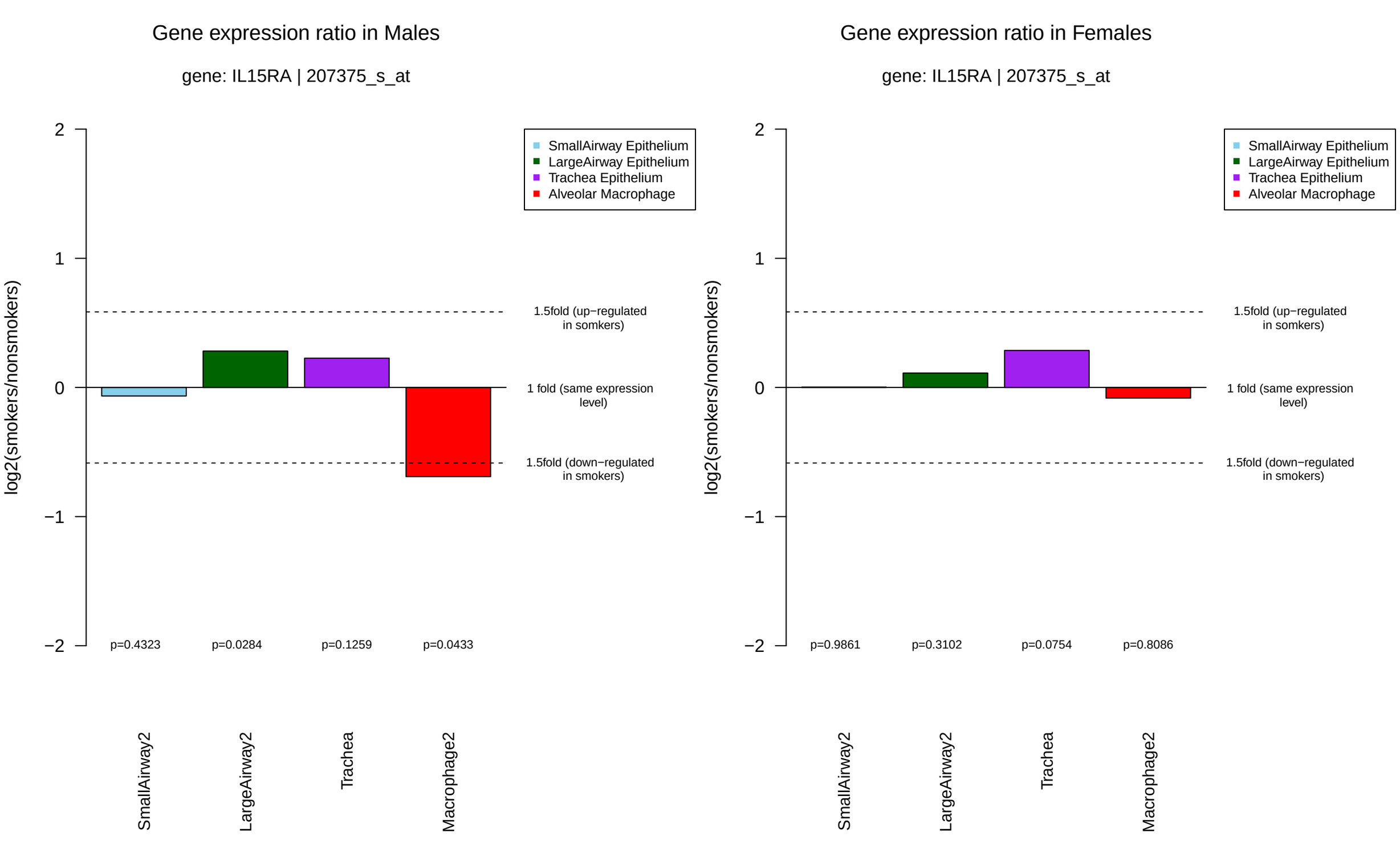
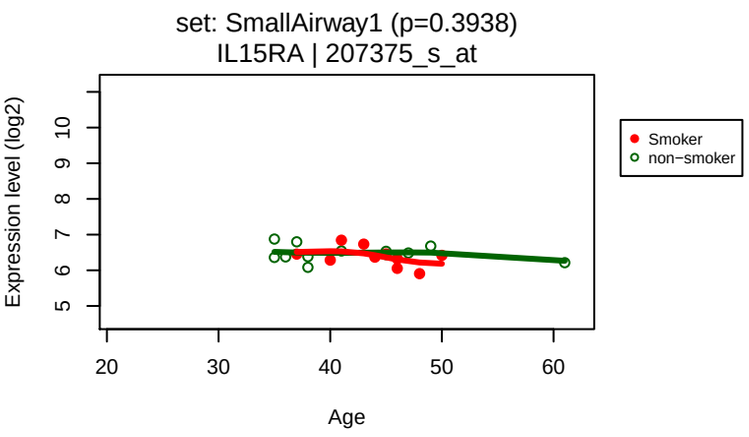
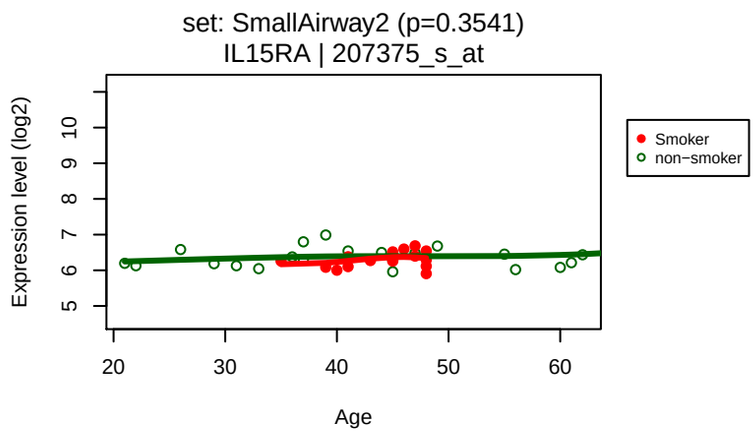
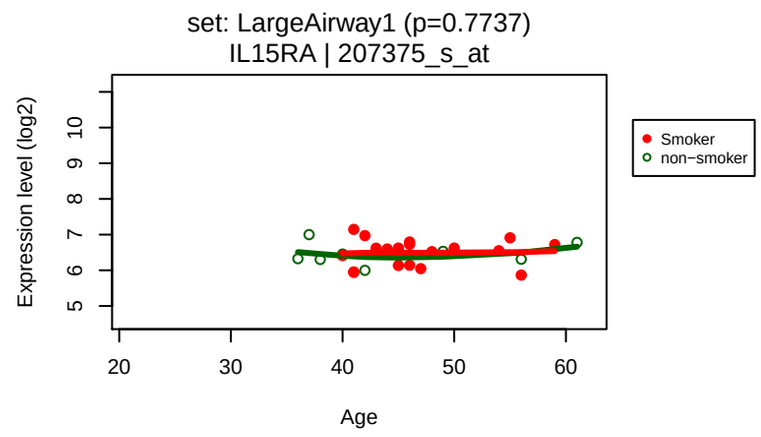

Gene expression of smokers and nonsmokers
gene: IL15RA | 207375_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3938
p=0.3541
p=0.5551
p=0.7737
p=0.0284
p=0.0444
p=0
p=0.0501
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

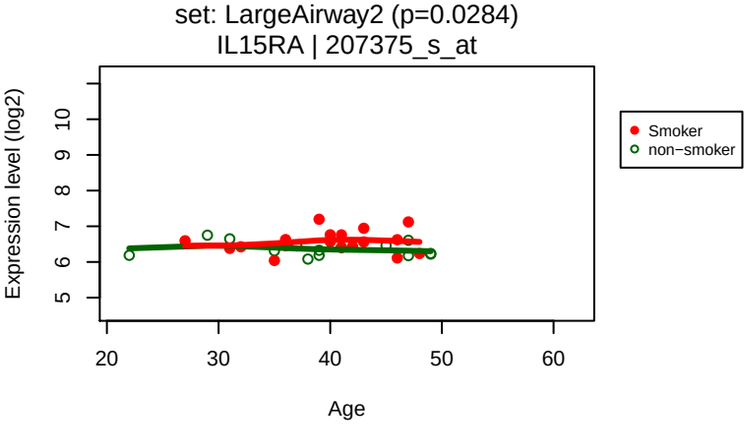

Gene expression ratio (smokers/nonsmokers)
gene: IL15RA | 207375_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3938
p=0.3541
p=0.5551
p=0.7737
p=0.0284
p=0.0444
p=0
p=0.0501
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

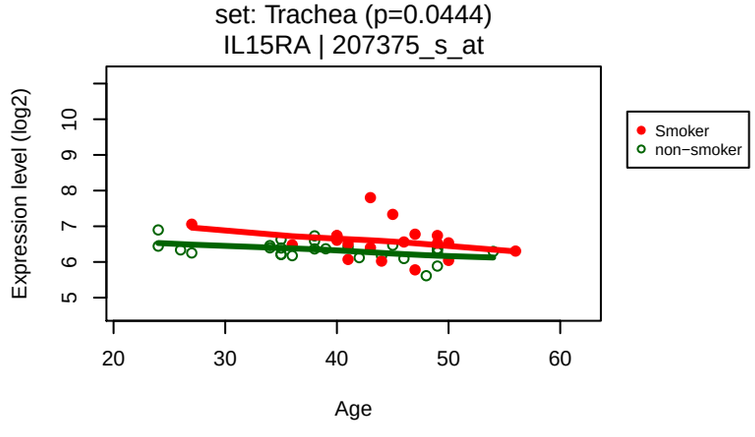

Gene expression ratio in Males
gene: IL15RA | 207375_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4323
p=0.0284
p=0.1259
p=0.0433
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IL15RA | 207375_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9861
p=0.3102
p=0.0754
p=0.8086
SmallAirway2
LargeAirway2
Trachea
Macrophage2

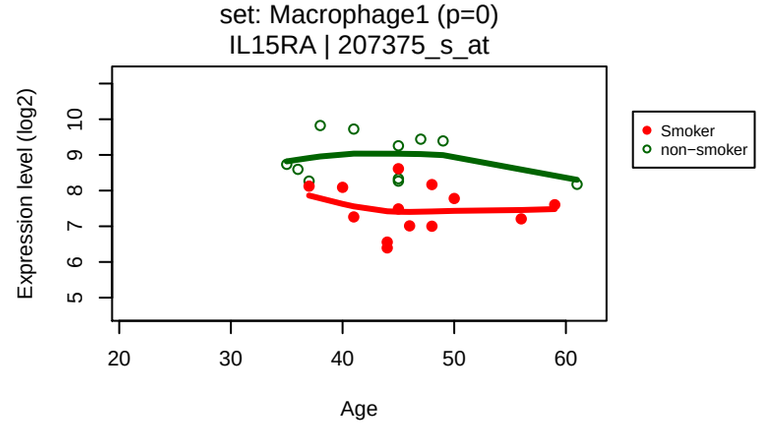

set: SmallAirway1 (p=0.3938)
IL15RA | 207375_s_at
Expression level (log2)
Age
Smoker
non−smoker

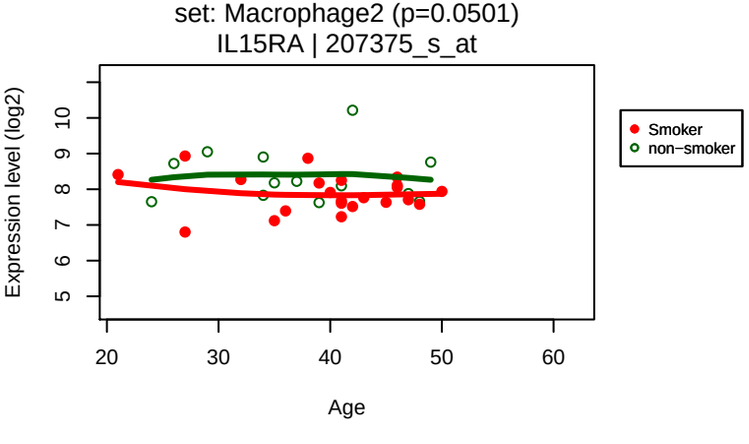

set: SmallAirway2 (p=0.3541)
IL15RA | 207375_s_at
Expression level (log2)
Age
Smoker
non−smoker

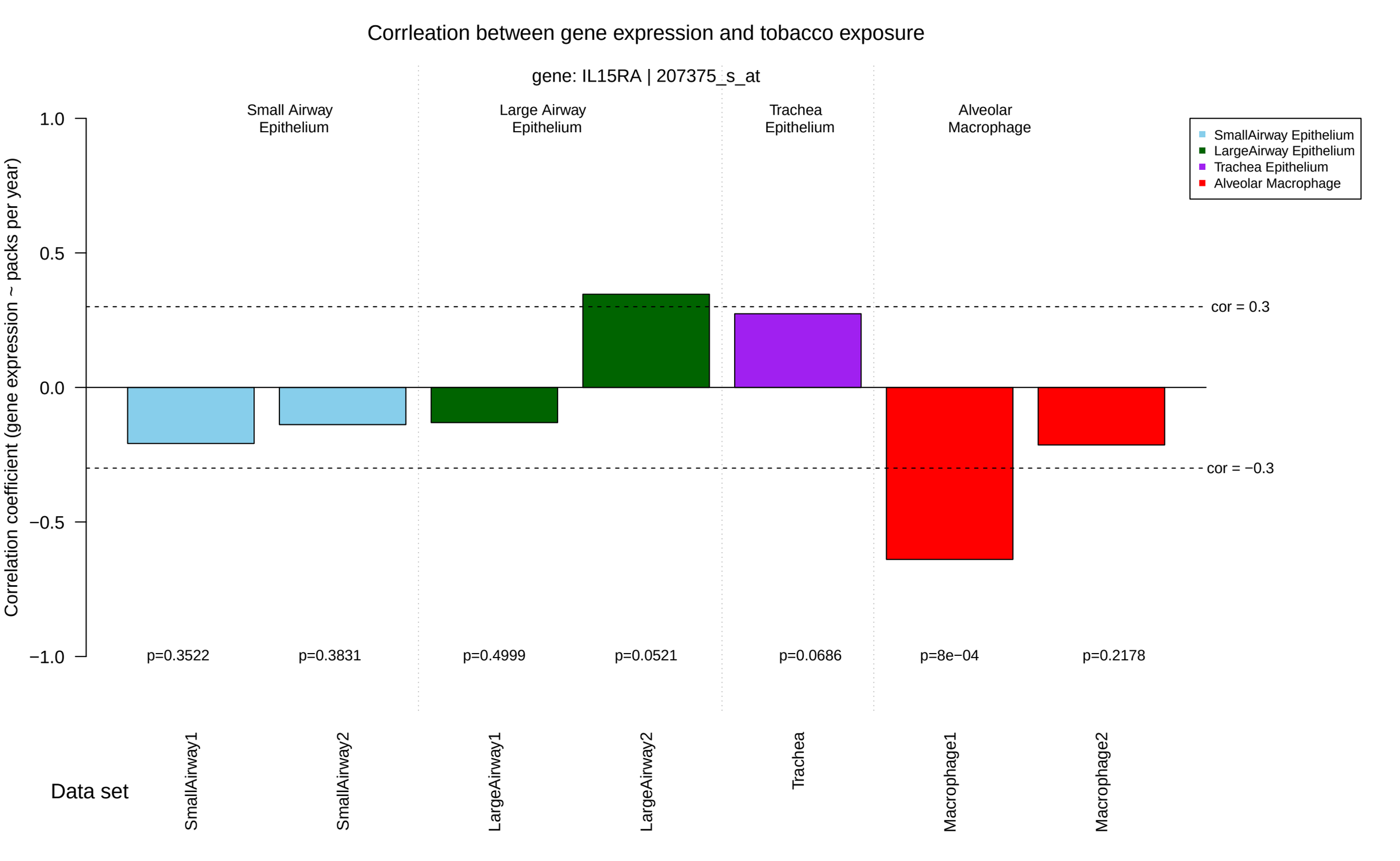

set: LargeAirway1 (p=0.7737)
IL15RA | 207375_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0284)
IL15RA | 207375_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0444)
IL15RA | 207375_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0)
IL15RA | 207375_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0501)
IL15RA | 207375_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IL15RA | 207375_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3522
p=0.3831
p=0.4999
p=0.0521
p=0.0686
p=8e−04
p=0.2178
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2